TEXORA · LECTURE SHEET

Introduction to Textile Testing

Testing turns opinion into numbers: why textiles are tested, what makes a test trustworthy, how samples are drawn and conditioned, and how results are reported so that a buyer, a spinner and a laboratory all mean the same thing by the same figure.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 1 · LEVEL 3 TERM 1

1. What textile testing is — and why it exists

Textile testing is the measurement of the properties of fibres, yarns, fabrics and garments by defined methods, on defined samples, under defined conditions, so that the result can be compared with a specification, with another lot, or with the same material tested elsewhere. A test that cannot be repeated by another laboratory to give nearly the same answer is not a measurement; it is an opinion with numbers attached.

Testing serves four distinct purposes, and confusing them leads to wrong sampling and wasted work:

- **Raw-material selection and pricing.** Cotton is bought on length, strength, fineness, trash and moisture; wool on diameter and yield. The test result moves money directly — a bale misjudged by one micronaire unit or 1 % regain is mispriced on every kilogram.
- **Process control.** Sliver weight, roving irregularity and yarn twist are checked so a drift is caught at the machine that caused it, not after a shift of yarn has been spun.
- **Product (quality) control.** Finished yarn and fabric are tested against the buyer's specification — count, strength, evenness, fastness — before despatch.
- **Research and development.** New fibres, blends and finishes can only be judged by controlled comparison, which is testing with an experimental design.

Two ideas run through the whole course. First, **textiles are variable**: no two cotton fibres in a bale are alike, so every result is an average of a sample and carries sampling error. Second, **textiles respond to the atmosphere**: fibres absorb moisture from the air, and moisture changes mass, dimensions, strength and electrical behaviour. Almost every precaution in testing — random sampling, replication, conditioning — exists to control one of these two facts.

2. Properties that are tested

GROUP	TYPICAL PROPERTIES	EXAMPLES IN THIS COURSE

Physical	Length, diameter, fineness, linear density, twist, crimp	Topics 3, 4, 5, 10, 11
Mechanical	Tensile strength, elongation, abrasion, bending	Topics 8, 12
Moisture relations	Moisture regain, moisture content, conditioning behaviour	Topic 2
Cleanliness / defects	Trash, neps, hairiness, imperfections	Topics 6, 7, 13, 14
Chemical	Fibre identification, damage, finish content, fastness	Covered with dyeing and finishing courses
Comfort / service	Air permeability, thermal resistance, handle	Fabric testing courses

3. Accuracy, precision and error

Accuracy is closeness to the true value; **precision** is closeness of repeated results to each other. A mis-calibrated balance can be precise and wrong. Errors divide into:

- **Systematic errors** — always in the same direction: a stretched tape, a balance reading 0.2 g high, testing unconditioned yarn on a humid day. They are removed by calibration and standard method, not by taking more readings.
- **Random errors** — vary from reading to reading: the genuine variation of the material and small fluctuations of technique. They are reduced (in the average) by replication: the standard error of a mean falls as 1/√n.
- **Gross errors (blunders)** — misread scales, wrong specimen, arithmetic slips. They are removed by checking, not by statistics.

$$\text{Standard error of the mean} \quad s_{\bar{x}} = s / \sqrt{n}$$

$$\text{Coefficient of variation} \quad CV\% = (s / \bar{x}) \times 100$$

Unless stated otherwise, results in this course are reported as the **mean** of the stated number of tests, with the CV% or the confidence limits when a specification decision depends on them.

WORKED EXAMPLE — HOW MANY TESTS ARE ENOUGH?

Single-yarn strength tests give a standard deviation s = 28 cN about a mean of 320 cN. (a) The standard error of the mean of 25 tests is 28 / √25 = 28 / 5 = **5.6 cN**, i.e. 1.75 % of the mean. (b) To halve that error to 2.8 cN requires √n = 28 / 2.8 = 10, so n = **100 tests** — quadrupling the work

> halves the error. This square-root law is why test numbers in standards (20, 50, 100 breaks) are compromises between confidence and laboratory time.

4. Sampling — the result can be no better than the sample

A laboratory tests grams from a consignment of tonnes. If those grams are not representative, no instrument can rescue the result. The governing principles are:

- **Random sampling.** Every unit in the lot must have an equal chance of selection. Taking fibres from the top of one bale is convenient sampling, and it is biased — bale surfaces differ from bale cores in moisture and trash.
- **Zoning.** For a tuft sample, small sub-samples are drawn from many zones of the bulk and combined, so local variation is averaged out.
- **Laboratory sample → test sample → specimen.** The bulk sample is reduced in stages (for example by quartering or by a sample divider), each stage preserving representativeness, until the small **specimen** that actually enters the instrument is reached.
- **Number of specimens.** Set by the variability of the property and the confidence required — see the worked example above. Fibre properties, which are highly variable, need hundreds to thousands of individual measurements; yarn lea tests need fewer, larger specimens.
- **Avoid damage and bias in handling.** Do not stretch, untwist, or preferentially lose short fibres or trash while preparing the sample; a fibre-length sample that has lost its short fibres will report a length the cotton never had.

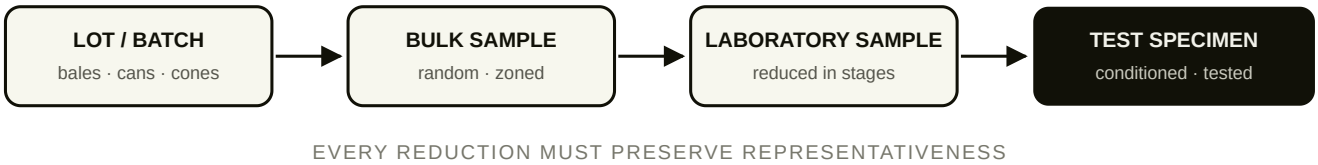


Figure 1 — The sampling chain. Most testing error is created here, before any instrument is switched on.

5. The standard atmosphere and conditioning

Because moisture changes almost every textile property, tests are made in a **standard atmosphere** and on specimens brought into moisture equilibrium with it:

STANDARD	TEMPERATURE	RELATIVE HUMIDITY
ISO / BS temperate standard atmosphere	20 ± 2 °C	65 ± 4 % RH

Tropical standard atmosphere (permitted alternative)	27 ± 2 °C	65 ± 4 % RH

Conditioning means exposing the specimen, freely, to the standard atmosphere until it reaches equilibrium — practically, until successive weighings at least 2 hours apart differ by less than 0.25 %. Specimens should approach equilibrium from the dry side where the standard requires it, because of **hysteresis**: a fibre drying down to 65 % RH holds more moisture than the same fibre wetting up to 65 % RH. Pre-conditioning at low humidity (for example 50 °C, 10–25 % RH) is used to set the dry-side approach. A **conditioned test** must be distinguished from a test made "as received" — the two will differ, and a report must say which was done.

6. Standards, calibration and the test report

- **Test standards** (ISO, ASTM, BS, BISFA) fix the method, apparatus, sampling, atmosphere, speed and calculation, so that "tested" means the same thing in Dhaka and in Manchester. Where a buyer names a standard, that standard — not local habit — governs.
- **Calibration** ties instruments to known references: standard weights for balances, gauge blocks and standard tapes for length, reference cottons for HVI, standard rubbers / standard fabrics where the method requires them. Calibration is recorded and repeated on schedule.
- **The report** states: material and lot identification; the standard followed; sampling and number of tests; atmosphere and conditioning; individual results or mean with CV% / confidence limits; any deviation from the standard. A single bare number with no method named is not a test report.

Glossary

Accuracy — closeness of a result to the true value. **Precision** — closeness of repeated results to each other. **Specimen** — the piece actually tested. **Conditioning** — bringing a specimen to moisture equilibrium with the standard atmosphere. **Standard atmosphere** — 20 ± 2 °C and 65 ± 4 % RH (tropical alternative 27 ± 2 °C). **Hysteresis** — the difference in moisture held at the same RH depending on whether the fibre arrived from wetter or drier conditions. **CV%** — standard deviation expressed as a percentage of the mean. **Calibration** — checking and adjusting an instrument against a known reference.

Key points

• Testing exists for raw-material pricing, process control, product control and research — each purpose dictates its own sampling.

• Textiles are variable and moisture-sensitive; sampling, replication and conditioning are the controls for those two facts.

• Systematic errors are removed by calibration and method; random errors shrink only as $1/\sqrt{n}$ in the mean.
• The sampling chain lot → bulk sample → laboratory sample → specimen must stay representative at every stage.
• Standard atmosphere: 20 ± 2 °C, 65 ± 4 % RH; condition to constant mass before testing.
• A report names the standard, the sampling, the number of tests and the spread — not just a mean.

Practice

1. A balance reads 0.3 g high on every weighing, however many specimens are averaged. Which class of error is this, and how is it removed?
2. Strength tests have s = 30 cN. How many tests are needed for a standard error of the mean of 3 cN?
3. A technician samples cotton only from the top layer of each bale because it is easy to reach. Name the fault and two properties whose results it may bias.
4. Why must a report state whether yarn was tested conditioned or as received? Refer to hysteresis in your answer.
5. Calculate the CV% of five lea strengths: 74, 79, 81, 77, 79 lb (use $s = \sqrt{[\Sigma(x-\bar{x})^2/(n-1)]}$).

Answer key

1. A **systematic** error. Averaging cannot remove it; it is removed by calibration against standard weights and correction of the balance.
2. $s_{\bar{x}} = s/\sqrt{n} \rightarrow 3 = 30/\sqrt{n} \rightarrow \sqrt{n} = 10 \rightarrow n =$ **100 tests**.
3. Convenience (non-random, biased) sampling. Bale surfaces differ from cores in **moisture regain** (exposure to air) and **trash / dust content** (settling and handling), so both results are biased; length may also be biased if surface tufts are damaged.
4. Moisture changes yarn mass and strength. Because of hysteresis, a specimen brought to 65 % RH from the wet side holds more moisture than one brought from the dry side, so "as received" results depend on the material's history. Only conditioned testing in the standard atmosphere, approached as the standard specifies, makes results comparable between laboratories.
5. Mean = (74+79+81+77+79)/5 = 390/5 = **78 lb**. Deviations: −4, +1, +3, −1, +1; squares 16+1+9+1+1 = 28; $s = \sqrt{(28/4)} = \sqrt{7} = 2.65$ lb. CV% = 2.65/78 × 100 ≈ **3.4 %**.